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| Course Code:<br>CSE228                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Course Title: Principles of Artificial Intelligence<br><br>Type of Course: Theory Only                                                                                                                                                                                                                                                                                                                                                                                                                           | L- T-<br>P- C | 3 | 0 | 0 | 3          |
| Version No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |   |   |   |            |
| Course Pre-requisites                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Mathematics: Logic, Algebra, Probability<br>Formal Languages                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |   |   |   |            |
| Anti-requisites                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | NIL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |   |   |   |            |
| Course Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>This Course will introduce the basic principles in artificial intelligence. It will cover representation schemes, problem solving paradigms, constraint propagation, search strategies, knowledge representation, Probabilistic Reasoning.</p> <p>Topics include: AI methodology and fundamentals, intelligent agents, search algorithms, game playing, supervised and unsupervised learning, uncertainty and probability theory, probabilistic reasoning in AI, Bayesian networks, statistical learning.</p> |               |   |   |   |            |
| Course Objective                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | The objective of the course is to familiarize the learners with the concepts of Principles of Artificial Intelligence and attain SKILL DEVELOPMENT through PARTICIPATIVE LEARNING techniques                                                                                                                                                                                                                                                                                                                     |               |   |   |   |            |
| Course Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>On successful completion of the course the students shall be able to:</p> <p>Explain the basic concepts of Artificial Intelligence.</p> <p>Apply techniques logic rules for Knowledge Representation.</p> <p>Apply Artificial Intelligence techniques for selected problem solving.</p> <p>Apply probabilistic reasoning in AI.</p>                                                                                                                                                                           |               |   |   |   |            |
| Course Content:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |   |   |   |            |
| Module 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Introduction to Artificial Intelligence and Knowledge based systems                                                                                                                                                                                                                                                                                                                                                                                                                                              | Comprehension |   |   |   | 9 Sessions |
| <p>Introduction to Artificial Intelligence, Definitions, foundation, History and Applications; Agents: Structure of Intelligent agent and its functions, reactive agents, deliberative agents, goal-driven agents, utility-driven agents, and learning agents; Introduction to Knowledge representation, approaches and issues in knowledge representation, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time, and space, Knowledge-based agent and its Structure, Knowledge-Based Systems; Frame Structures, Conceptual graphs.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |   |   |   |            |

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| Module 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Logic based Knowledge Representation       | Application |  | 9 Sessions  |
| Introduction, Syntax and Semantics, Proof Systems, Natural Deduction, Tableau Method, Resolution Method, Propositional Logic, Predicate Logic, First order Logic, Properties of well-formed formulas (Wffs), Conversion to Clausal Form, The Resolution Principle, Inference in First Order Logic (FOL)                                                                                                                                                                                                                                                                              |                                            |             |  |             |
| Module 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Problem Solving by searching               | Application |  | 12 Sessions |
| Introduction to Problem space and state space, State space search techniques solving problems by searching: forward and backward, state-space, blind, heuristic, problem-reduction, A, A*, AO*, minimax, constraint propagation, neural, stochastic, and evolutionary search algorithms, sample applications, Introduction to reasoning, various types of reasoning methods, Certainty factors and rule-based systems Dempster Shafer Theory.                                                                                                                                        |                                            |             |  |             |
| Module 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Learning and Probabilistic reasoning in AI | Application |  | 10 Sessions |
| Introduction to learning, Forms of Learning: Statistical learning, Supervised Learning, Unsupervised Learning, Learning rules of AI, Probabilistic reasoning in AI, Bayesian networks, Hidden Markov Model.                                                                                                                                                                                                                                                                                                                                                                          |                                            |             |  |             |
| Targeted Application & Tools that can be used:<br>Google Colab, Python                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                            |             |  |             |
| Text Book<br><br>Stuart J. Russell and Peter Norvig, Artificial intelligence: A Modern Approach, 3rd edition, Upper Saddle River, Prentice Hall.<br><br>Elaine Rich, Kevin Knight and Shivashankar B. Nair, "Artificial Intelligence", TataMcGraw-Hill, Third Edition, 2009[R.N.].                                                                                                                                                                                                                                                                                                   |                                            |             |  |             |
| References<br><br>1. N J Nilsson (1997). Artificial Intelligence- A new synthesis, Elsevier Publications.<br>2. N J Nilsson (1982). Principles of Artificial Intelligence, Springer.<br>3. Patterson, D. W. (1990). Introduction to artificial intelligence and expert systems. Englewood Cliffs, Prentice Hall.<br>4. Luger, G. F. (2002). Artificial intelligence: Structures and strategies for complex problem solving, Harlow, Pearson Education.<br><br>E-Resources<br><br><a href="https://puuniversity.informaticsglobal.com">https://puuniversity.informaticsglobal.com</a> |                                            |             |  |             |
| Topics relevant to "SKILL DEVELOPMENT": Knowledge Based Systems, Probabilistic reasoning in AI, Bayesian networks, Hidden Markov Model for Skill Development through Participative Learning techniques. This is attained through the assessment component mentioned in the course handout.                                                                                                                                                                                                                                                                                           |                                            |             |  |             |

